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Conventional methods of dissolved gas analysis using oil-immersed power transformer for fault diagnosis: A review

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ABSTRACT

This review paper summarizes the discoveries made about dissolved gas analysis (DGA) conventional methods around the decade. DGA is a well-known diagnostic tool to classify different fault types that are likely to occur in oil-filled transformers. Interpretation methods for fault detection include the Key Gases Method, Duval Triangle Method, Duval Pentagon Method, and Ratio Methods consisting of the IEC Method, Rogers Method, and Doernenburg Method. In this review paper, a brief introduction of DGA will be given, followed by a review of each method, and the correlation of fault types between methods has also been reviewed so that the readers have a clear view of the capability of each method. The paper highlights the merits and demerits of each method that can be guided before selecting the most suitable DGA method for performing the diagnostic analysis. In this regard, it is of utmost importance to enhance the reliability and accuracy of each DGA method in the diagnostic process, since it detects early transformer faults, and therefore prevents further damage to the specific transformer. Finally, the conclusion and future works on this topic also will be presented accordingly.

1. Introduction

Power transformer is one of the utmost important and expensive pieces of equipment for power system utility [1]. It converts electrical energy from lower to higher voltage or vice versa via a magnetic flux medium that transfers electrical energy from one circuit to another. There are two types of insulating mediums for transformers which are: dry type and liquid immersed type depending on their application. For instance, a dry type insulating medium is more appropriate for indoor use because it is normally air-cooled and has category A, B, C, or H winding insulation. Thus, it can be either self-cooled or forced air-cooled [1]. Whereas liquid immersed type is commonly used outdoors. As such, it has the benefits of being smaller in size, less expensive, and has improved overload capabilities [2,3].

Power transformers are important components of the power grid both in transmission and distribution systems. Thus, it is important to ensure that this equipment is managed, controlled, and maintained properly for long-term use. Moreover, early detection of transformer faults can help to avoid interruption of service, abnormal operating conditions, and undesirable service loss [4,5]. This is because it requires expensive repairs and prolonged downtime if the failure occurs. When a fault occurs, such as overheating failure, partial discharge (corona), or arc discharge (arcing), the insulating medium is exposed to abnormal electrical or thermal stresses. For instance, the insulation breakdown is initially caused by internal arcing. In this state, the transformer will be disconnected and will embrace the affected winding [6].

The IEEE committee on power transformers studied in 2009 that due to an overpressure phenomenon inside the transformer, the transformer tank will explode or distort. It is commonly due to the generation of a large volume of decomposed gas caused by an internal arcing fault. Subsequently affecting the dynamic pressure and static overpressure wave to exceed the mechanical strength of the transformer tank [7]. These faults also lead to the degradation of the insulating medium and releasing byproducts such as Methane (CH_4), Acetylene (C_2H_2), Ethylene (C_2H_4), Ethane (C_2H_6), Oxygen (O_2), Carbon Dioxide (CO_2), Carbon Monoxide (CO), Hydrogen (H_2), and Nitrogen (N_2) [8–13]. The composition of the by-product gases can be used to differentiate between various types of faults. For example, internal faults emitted the H_2, CH_4, C_2H_6, and C_2H_4 gases, while the cellulose faults emitted CH_4, H_2, CO, and CO_2 gases [14–17].

Several studies have been performed on monitoring and diagnostic

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Abbreviations

CO	Carbon Oxide
CDP	Combined Duval Pentagon
DPM	Duval Pentagon Method
DGA	Dissolved Gas Analysis
DRM	Doernenburg Ratio Method
DTM	Duval Triangle Method
FID	Flame Ionization Detector
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
KGM	Key Gases Method
MPM	Mansour Pentagon Method
OFAF	Oil Forced Air Force
ONAF	Oil Natural Air Forced
RM	Ratio Methods
RRM	Rogers Ratio Method
PD	Partial Discharge
TCD	Thermal Conductivity Detector
TDCG	Total Dissolved Combustible Gas
TFE	Total Fault Energy

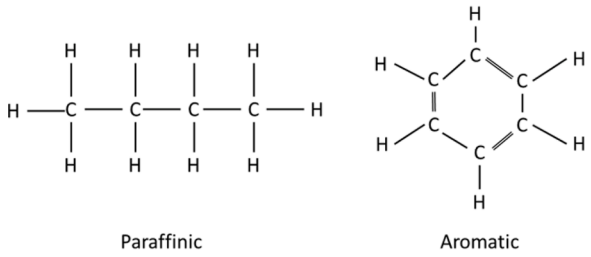


Fig. 1. Molecular structure of hydrocarbons [2].

procedures to determine the transformer's health in order to utilize preventive and maintenance measures. Dissolved Gas Analysis (DGA) is one of the most widely accepted procedures for transformer health diagnosis [11,18-21]. The valuable feature of the DGA approach is that it can discover emerging faults early, track fault development rates, validate fault presence, as well as provide a repair schedule. The only drawback of the DGA techniques is the disability to determine the transformer faults for the cases that are out of the specified codes thus providing poor diagnostic accuracy [22,23]. Photoacoustic spectroscopy or gas chromatography technique are applied to extract the dissolved gases of transformer sample oil [24]. Based on the extracted data, DGA is utilized to identify the type of fault using a different approach such as Key Gases, Duval Triangle, Duval Pentagon, and Ratio Methods (Doernenburg, Rogers, and IEC).

There are several artificial intelligence-based techniques such as artificial neural networks, adaptive neuro-fuzzy inference systems, fuzzy logic, expert systems, or hybrid system have been suggested for transformer fault diagnosis [25–28]. However, these techniques are not suitable for a wide range of practical implementations due to the dependence on the reliability of the training data set and the requirement for repetitive training for a new set of training data. The improper training process can affect the accuracy of the fault diagnosis or to lose its generalization.

In this comprehensive review, the health of the transformer is evaluated using the previous well-known conventional DGA methods mentioned above. Different values of gas concentration are used including H_2, CH_4, C_2H_2, C_2H_4, and C_2H_6 gases to observe the accuracy and reliability of each method. Then, the performance of each method is analysed and compared with its own set of advantages and disadvantages in which their accuracy and reliability in interpreting transformer conditions are still questionable [18].

2. Mineral oil decomposition

Mineral oil, which is made from petroleum, is typical liquid insulation and cooling agent medium for liquid-filled power transformers. Mineral oil is preferred due to its satisfactory dielectric properties, which include high electric strength, low viscosity, low fire point, and low pour point. As represented in Fig. 1, mineral oils are a mixture of hydrocarbon molecules such as CH, CH_2, and CH_3 in the shape of aromatic or paraffinic compounds [2,29]. Mineral oil will decompose and release fault gases which are H_2, CH_4, C_2H_2, C_2H_4, and C_2H_6 when a fault occurred in the transformer.

The decomposition in mineral oil can be categorized into two, which are cellulose decomposition and oil decomposition. Cellulose decomposition is a thermal degradation of cellulose that involves three basic mechanisms involving hydrolysis, oxidation, and pyrolysis [8]. These mechanisms release CO and CO_2, as well as H_2 and CH_4 [11]. Mineral oil is made up of a combination of hydrocarbon molecules with a general formula of C_nH_{2n+2}, with n ranging from 20 to 40. The fault gases form when carbon-hydrogen and carbon-carbon bonds collapse due to the oil decomposition of thermal and electrical faults. These free radicals will combine H_2, CH_4, and C_2H_6 as well resulting in the formation of C_2H_4 and C_2H_2 if the mineral oils undergo further decomposition.

Fig. 2 illustrates the evolution of gases extracted from oil as a function of the breakdown energy. At a low temperature of 150 °C, H_2 and CH_4 begin to form in relatively small amounts. At approximately 250 °C, C_2H_6 production begins and eventually exceeds CH_4 at 275 °C. At temperatures under 300 °C, detecting hot spots is extremely challenging. Furthermore, when decomposition occurs when 200 °C and 300 °C, the formation of CH_4 exceeds H_2. At about 350 °C, C_2H_4 begins to form and C_2H_2 begins to form between 500 - 700 °C later. When the temperature is above 700 °C, a thermal fault or hot spot can produce small amounts of C_2H_2, while an internal arcing fault can produce higher amounts of C_2H_2. H_2 production exceeds as much as the other gases within about 450 °C until almost 750–800 °C when more C_2H_2 is formed [30].

The main causes of the gas formation within insulation oils are thermal faults and electrical faults. Thermal decomposition of mineral oil occurs when low molecular weight gases (H_2, CH_4) are released as a result of a thermal fault, as well as high molecular weight gases (C_2H_6, C_2H_4) are released at temperatures 150 to 500 °C [20]. If the temperature of the mineral oil decomposition exceeds 500 °C, then a large amount of C_2H_2 will be released. Meanwhile, thermal decomposition in paper insulation or other solid insulation will release CO_2 and CO gasses [31]. An electrical fault is essentially determined by the amount of energy dissipated by partial discharge and arcing. Partial discharge occurs when a solid or liquid is pressured by high voltage and the dielectric breakdown of some of its parts occurs. Due to high voltage stresses the mineral oil, will produce several gasses, cold plasma discharges, bubbles, or voids, plus X-wax will be formed in the paper insulation. Arcing is a high-energy discharge in which gas insulation failure produces continuous plasma discharge as a result of current flowing through a non-conductive medium such as air [20]. The arc fault is the most serious fault because it generates a high volume of C_2H_2 gas compared to partial discharge which is a less serious fault that only produces a large amount of H_2 and CH_4 gasses.

3. DGA as diagnostic tools

DGA is a reliable diagnosis method to detect incipient failures when insulating oils are exposed to abnormal electrical or thermal stresses. During the event, the insulating oils will degrade and release small amounts of gases. The composition of the released gases can be used to identify different types of faults such as partial discharge, arc discharge,

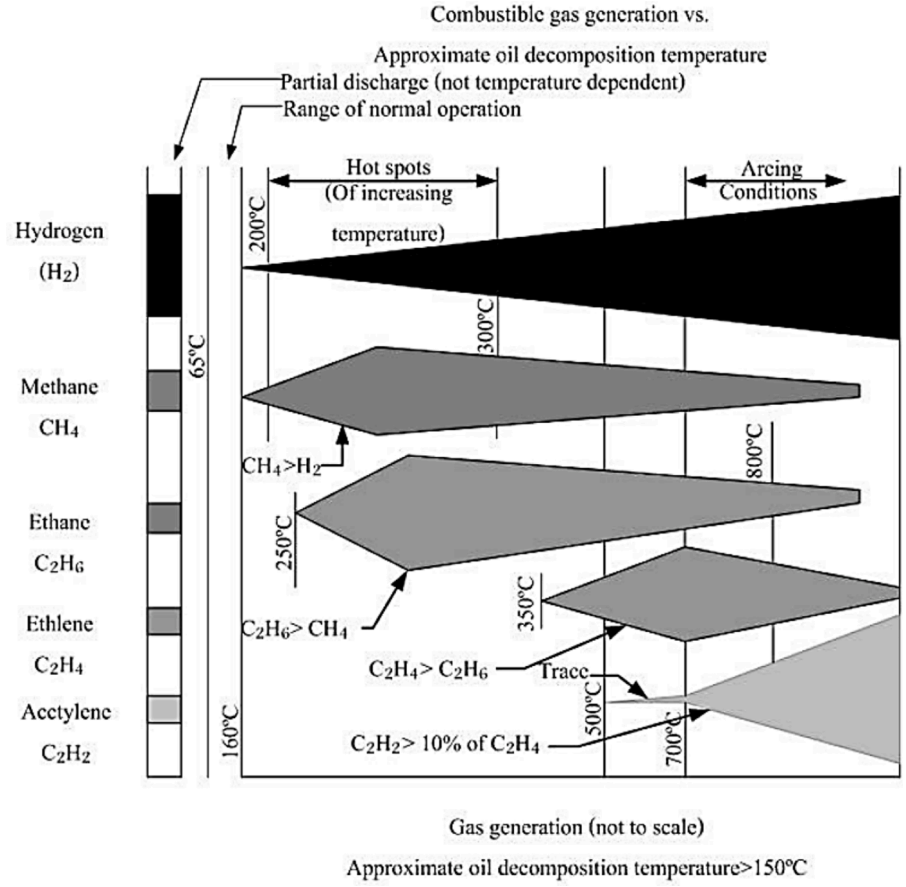


Fig. 2. Relative rates of gas evolution in a form of decomposition energy [30].

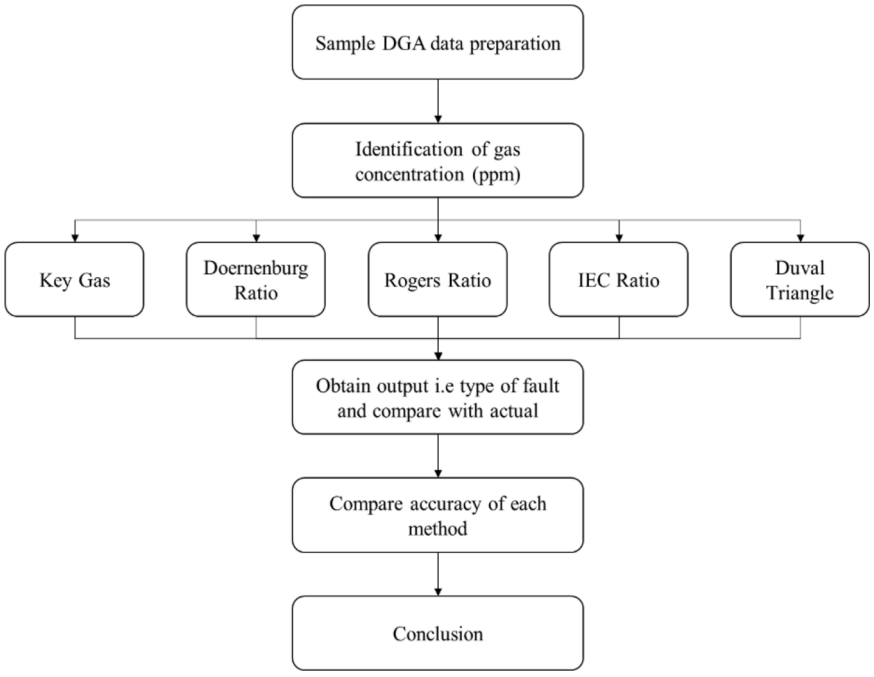


Fig. 3. DGA method classification.

Table 1
Fault type based on key gas.

Key Gas	Fault Type
C_2H_4	Thermal oil
CO	Thermal cellulose
H_2	Partial discharge
C_2H_2, H_2	Arcing

Table 2
Limit / Status concentration for each gas [33–35].

Gas Parameters	Status			
	Normal	Caution	Abnormal	Danger
H_2	100	101 - 700	701 - 1800	> 1800
CH_4	120	121 - 400	401 - 1000	> 1000
C_2H_2	1	2 - 9	10 - 35	> 35
C_2H_4	50	51 - 100	101 - 200	> 200
C_2H_6	65	66 - 100	101 - 150	> 150
CO	350	351 - 570	571 - 1400	> 1400
CO_2	2500	2501 - 4000	4001 - 10,000	> 10,000
TDCG	720	721 - 1920	1921 - 4630	> 4630

Table 3
Enthalpy value for each fault gas [32].

Gas	$\Delta H^\circ_{reaction}$ (kJ/mol)
H_2	97.9
CH_4	35.1
C_2H_2	268
C_2H_4	93.5
C_2H_6	57
CO	101.7
CO_2	283

and overheating. There are a few advantages of DGA including forewarning of developing faults, monitoring the rate of fault development, confirming the presence of faults, convenient repair scheduling, and affordability to checking the condition during an overload. DGA involves a few steps as mentioned as follows [21]:

1 Take a sample of oil from the unit.
2 The sample oil is extracted and subjected to mass spectroscopy to separate the dissolved gases from the oil.
3 Identify the occurrence of faults by detecting gas concentration levels using portable gas chromatographs such as FID or TCD. The concentration of dissolved gas is expressed in the microliters per litre (μL/L) also known as parts per million by volume (ppm v/v).
4 Different types of faults are discriminated by analysing the extracted data using suitable diagnostic methods such as KGM, IEC, RRM, DRM, and DTM.

Fig. 3 illustrated the flowchart of the DGA analysis. First, the input data of gas concentrations in parts per million (ppm) is extracted. Then, the extracted data are applied to different types of identification methods. Finally, the accuracy and reliability of each method are evaluated. A brief explanation for each method is described as follows.

4. Key gases method (KGM)

The number of fault gases emitted from the insulating oils determines the fault type using the Key Gases Method (KGM). The gas emission can be classified into hydrocarbon and hydrogen (C_2H_2, CH_4, C_2H_6, C_2H_4, H_2), non-fault gases (O_2, N_2), and carbon oxides (CO_2, CO). KGM monitors the fault based on the presence and relative proportions of each key gas. When the chemical structure breaks down at different levels of temperatures, the faults such as thermal failures in the cellulose and oil, partial discharge, and arcing will have occurred. Table 1 summarizes the key gases produced due to a specific type of fault.

It is commonly accepted that C_2H_4 is produced from low-temperature thermal oil faults with a small amount of CH_4, C_2H_6, and H_2 gases, while a small amount of C_2H_2 is traced at very high temperatures. During the occurrence of a thermal cellulose fault, the most common gas of CO is produced, and also a small amount of hydrocarbon gasses is detected. H_2 gas appears to be a major element to be traced in insulation during the occurrence of high-temperature thermal heating of

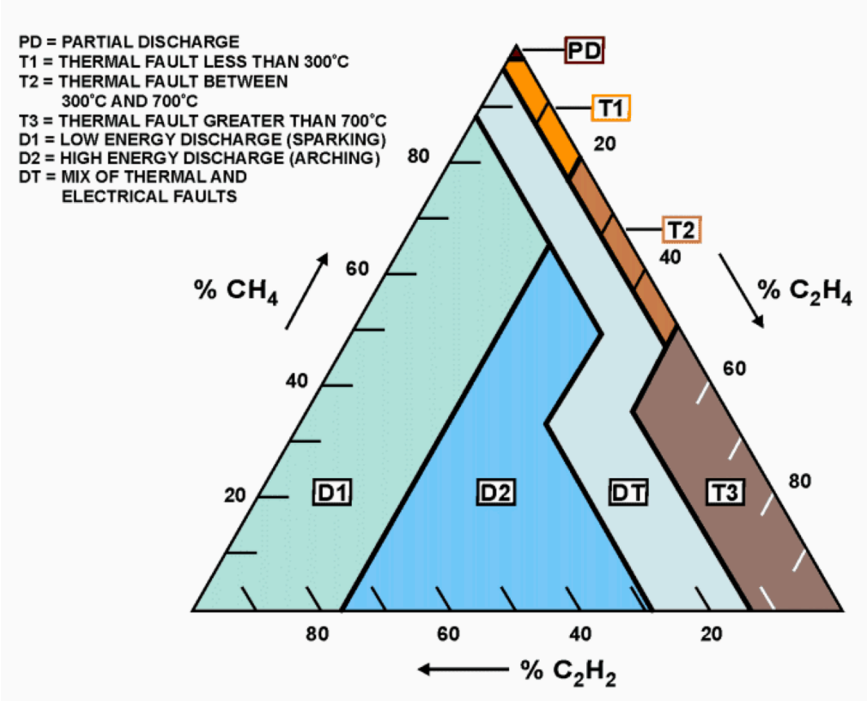


Fig. 4. Coordinates and fault zones in DTM [40].

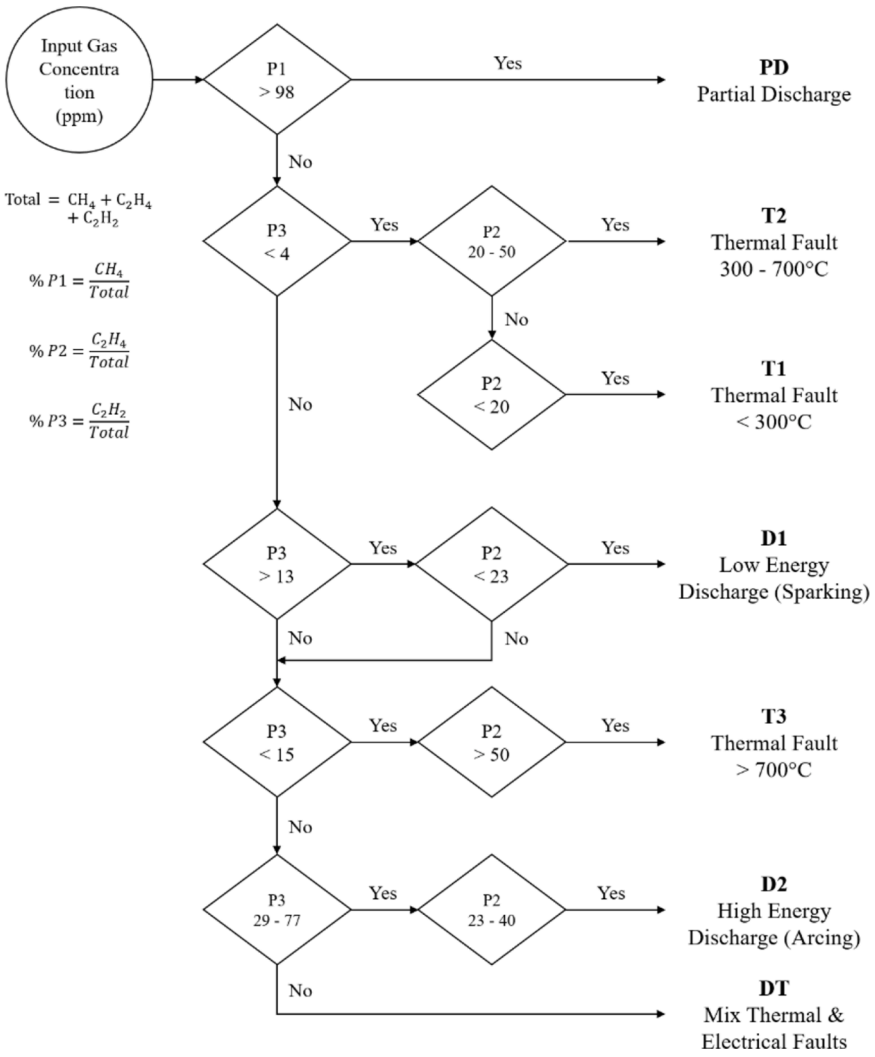


Fig. 5. Flowchart for DTM fault classification [30].

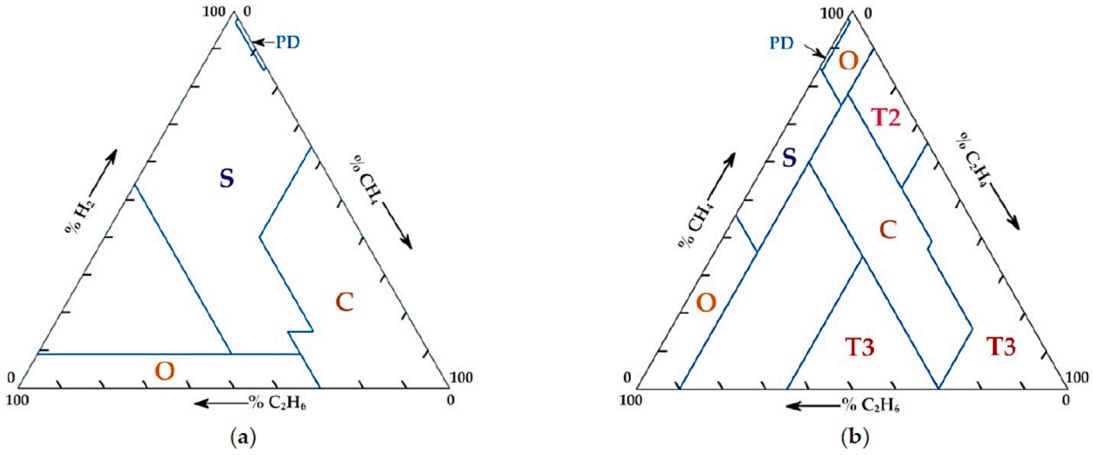


Fig. 6. (a) Duval4 for a low-temperature fault (b) Duval5 for a high-temperature fault [44].

partial discharge. Due to partial discharge, small amounts of CH_4, C_2H_6, and C_2H_4 are also traceable. Unlike the other faults, the arcing fault in the insulation oil produced 2 predominance gasses which are C_2H_2 and H_2 at very high temperatures. A small amount of CH_4, C_2H_6, and C_2H_4 are also traceable. But if arcing occurs in cellulose insulation, only CO is released.

It is noted that KGM discriminates the fault based on the highest number of key gas concentrations. This led to serious drawbacks because it is not always an obvious indicator to determine which gas produced a high concentration. Furthermore, CO concentration is not always a reliable indicator to differentiate the fault [31]. The amount of gas and its type are determined by the level of fault in the transformer, as well as

Table 4
Fault area coding.

Code	Fault Type
PD	Partial discharge
S	Stray gassing of mineral oil
C	Paper carbonization is caused by hot spots ($T > 300$ °C)
O	Overheating ($T < 250$ °C)
T2	Thermal faults are caused by the temperature of 300 °C to 700 °C
T3	Thermal faults are caused by an extremely high temperature at ($T > 700$ °C)

the degree and energy of the issue. There are five chemical reactions involved in the transformer fault diagnosis approach using the non-instructive DGA method. For example, from Eicosane ($C_{20}H_{42}$) molecule, the decomposition of mineral oil contributes to various gas emissions inside the transformer. Chemical reactions can be expressed as follows [32]:

$$C_{20}H_{42}\ (l) = CH_4\ (g) + C_{19}H_{38}\ (l) \quad (1)$$

$$C_{20}H_{42}\ (l) = H_2\ (g) + C_{20}H_{40}\ (l) \quad (2)$$

$$C_{20}H_{42}\ (l) = C_2H_2\ (g) + H_2\ (g) + C_{18}H_{38}\ (l) \quad (3)$$

$$C_{20}H_{42}\ (l) = C_2H_4\ (g) + C_{18}H_{38}\ (l) \quad (4)$$

$$C_{20}H_{42}\ (l) = C_2H_6\ (g) + C_{19}H_{36}\ (l) \quad (5)$$

where (l) represents liquid and (g) represents gas.

The IEEE C57.104–2008 standard provides a detailed guide for performing DGA testing. The purpose of this guideline is to provide information about the health of transformers based on standard default limit values. The limitation of default gas values is shown in Table 2.

Mharakurwa et al. [15] stated that to release the fault gases, a specific amount of energy is required. Furthermore, each different gas and amount of energy will determine the severity of the faults. As a result, the faulty energy is estimated using the reaction's enthalpy change ($\Delta H^{\circ}_{reaction}$) and the equation is written as follows:

$$\Delta H^{\circ}_{reaction} = \Delta H^{\circ}_{f(products)} - \Delta H^{\circ}_{f(reactants)} \quad (6)$$

where $\Delta H^{\circ}_{reaction}$ is the enthalpy of product formation.

Table 3 shows the calculated enthalpy for each gas.

The total fault energy (T.F.E) was calculated using the dissolved gas analysis equation as follows:

$$|\text{T.F.E.}| = \left(\frac{273+\theta}{273 \times 22400}\right)\left[\sum_{i=1}^{n}\left(\Delta H_{r,i} \times c_i\right)\right] \quad (7)$$

where $\Delta H_{r,i}$ is the calculated enthalpy during the fault gas reaction and c_i is the concentration of current gas in ppm.

Based on Eq. (7), different types of energy such as paper thermal fault energy, oil thermal fault energy, arcing fault energy, partial discharge fault energy, and total fault stress energy were calculated based on the key fault gases approach. The calculated energy also depends on the temperature of the oil sampling. With the combination of fuzzy logic diagnostic tools, the severity labels known as ***LOW, MEDIUM***, and ***HIGH*** can be classified based on the lower and upper limits of the fault energy.

Shanker et al. [12] have applied the KGM method to conduct a DGA test on two different transformers in a thermal power station in India. Subsequently, the respective proportion of each total dissolved combustible gas (TDCG) in the tested oil is plotted. Each fault type has a different pattern in the histogram which is characterized by key gases. Based on the observation, H_2 is the most abundant gas in the oil (relative proportion above 80%), which indicates the occurrence of electrical partial discharge fault. Meanwhile, C_2H_4 is the most abundant gas in relative proportion around 80%, which implies the presence of a thermal oil fault.

Whereas Syafruddin and the group [36] have measured the possibility of default abnormalities in power transformers (GT 2.1, ONAF) according to IEEE std C57.104 and IEC 60,599. As stated in IEEE std C57.104, the overheated oil faults should obtain a relative proportion of C_2H_4 around 63%, overheated cellulose fault should obtain a relative proportion of CO gas around 92%, partial discharge fault should obtain a relative proportion of H_2 around 85%, and arcing fault should obtain a relative proportion of H_2 and C_2H_2 around 60% and 30%, respectively. From their KGM analysis, it was observed that the presence of CO gas in the transformer is abnormal, in which the relative proportion of CO gas in Sample 1 (28/06/2018) and Sample 2 (18/12/2018) are 69.6% and 91.1%, respectively, indicating the thermal cellulose default of the transformer.

Irwanto [37] has conducted a key gas analysis to detect the amount of flammable gas in transformer oil at PT. Krakatau, Serang Banten, Indonesia. The data obtained are then compared with the limit of gas concentration stated under the IEEE C57.104.2008 standard and with certain conditions as reported in [37]. Based on their observation, CO gas has a concentration of 53 and a percentage of 76% which exceeded the gas limits. In addition to that, the obtained data indicate that the dominant CO gas causes excessive heat in paper insulation or overheated cellulose.

5. Duval triangle method (DTM)

Instead of classifying the fault based on the type of gasses produced, the Duval Triangle Method (DTM) which has been proposed by Michel Duval during the early 1970s has exploited the relative percentage of three combustible gases which are CH_4, C_2H_2, and C_2H_4 to classify different types of faults [33, 35]. The selection of these gases is based on the energy level required to generate the gases in transformers as described below [38]:

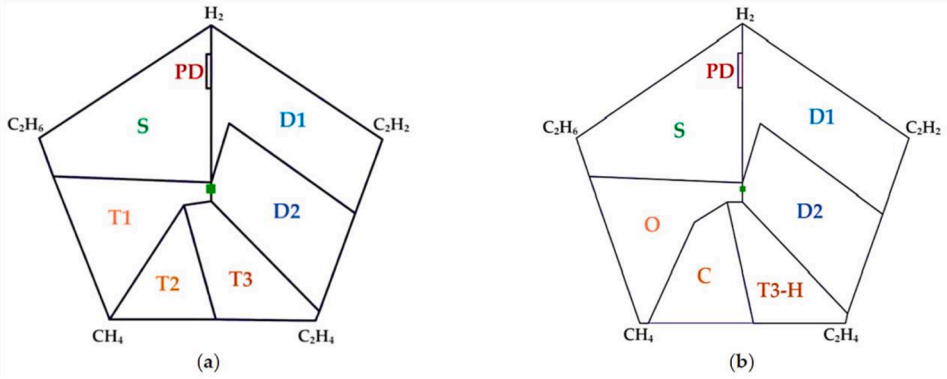


Fig. 7. Duval Pentagon to represent fault areas: (a) Pent1; (b) Pent2 [42].

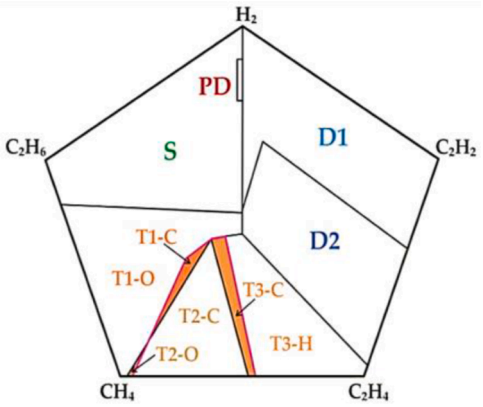


Fig. 8. Combined Duval Pentagon with 10 fault zones [47].

i CH_4: to detect low-energy or temperature faults
ii C_2H_4: to detect high-temperature faults
iii C_2H_2: to detect very high-energy or temperature or arcing faults

Generally, DTM is a diagram approach used to identify the fault types in the DGA of oil-filled high-voltage transformers. In this approach, the proportionate concentration of the mentioned gases is utilized as point coordination in an equilateral triangle thus categorizing the fault zone. It is observed that each point in the equilateral triangle relates to a unique combination of gas quantities [39]. To simplify the determination of fault type, Fig. 4 illustrates the relative percentages of these three gases which are plotted on each side of the triangle to determine partial discharge, thermal fault, and electrical fault [40,41]. The thermal fault is further comprised of hot spots of various temperature ranges, whereas the electrical fault consists of high- and low-energy arcing.

Generally, DTM is a 'closed system' diagnosis method which able to produce a result regardless of the availability of all gases. Therefore, the proper and careful handling of the method is required to avoid errors and incorrect results [30]. Fig. 5 shows a flow chart for DTM fault type classification. DTM shows a relatively efficient to classify the fault types that occurred in the mineral oil-filled transformer. However, it faces difficulty if the DGA results lie on the boundary of two fault areas that are unable to determine the actual fault. Besides that, DTM is unable to differentiate whether there is a fault or not in the transformer since there is no allocated area for the "No-Fault" case. The whole area of the triangle is dedicated to differentiating various fault types based on the percentage concentration of CH_4, C_2H_2, and C_2H_4 gasses.

In addition, DTM is only applicable when the fault condition has been diagnosed. Therefore, it cannot be used to identify changes from the normal condition to fault state. It cannot also be used to differentiate the condition of normal insulation ageing and fault. Furthermore, DTM is unable to be applied to discriminate the fault when the measurement of gas-in-oil concentration is taken at a low concentration. For example, a relatively high concentration of CH_4, as well as undetectable C_2H_2 and C_2H_4, were observed during low-energy degradation, indicating the occurrence of partial discharge, which was not always accurate. Therefore, it will give the comparative inaccuracy identification except when the gas concentrations are significantly above the detection limit. Besides that, it was observed that some oils tend to produce low-temperature gases between 80 °C to 200 °C which may cause incorrect fault identification in the transformers [42–44].

Table 5
Thermal fault areas coding.

Code CDP	Pent1	Pent2	Fault Description
T1 - O	T1	O	Thermal problem with no carbonization of the solid insulation (predicted temperature < 300 °C)
T1 - C	T1	C	Thermal problem with probable paper implication indicates carbonization (predicted temperature < 300 °C)
T2 - O	T2	O	Thermal problems are unlikely to involve solid insulation or carbonization of the paper (temperature 300 - 700 °C)
T2 - C	T2	C	Thermal problem with a high probability of paper implication (Probability of 80%, temperature 300 - 700 °C)
T2 - C	T2	C	Thermal emission only in the oil (temperature > 700 °C)
T3 - H	T3	T3 - H	Thermal problem with paper implication in the occurrence of a fault (carbonization) (high temperatures, T > 700 °C)

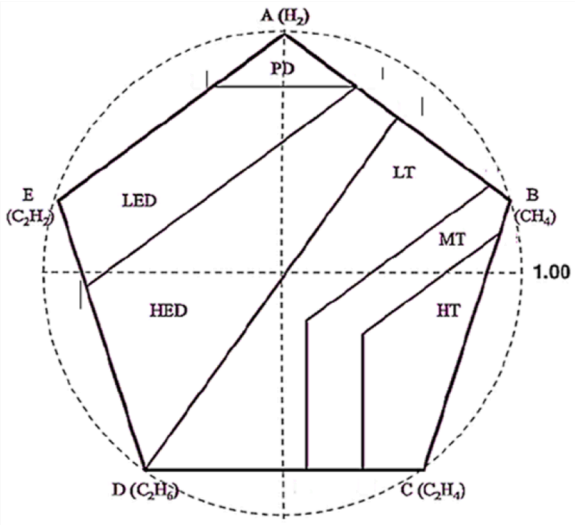


Fig. 9. Mansour Pentagon Method [39].

Therefore, to overcome all the mentioned drawbacks, Michel Duval has proposed other new triangles, which are Duval Triangle 4 (Duval4) and Duval Triangle 5 (Duval5). Duval4 is dedicated to low-temperature faults, while Duval5 is specified for high-temperature thermal faults, as shown in Fig. 6 [44]. It is important to note that Duval4 and Duval5 act as complementary triangles to the DTM and cannot be used to identify electrical faults. Table 4 defined all the fault areas used in Duval4 and Duval5. As mentioned previously, after the fault has been identified by DTM, subsequently Duval4 and Duval5 are applied to obtain more information to verify the fault that is accompanied by uncertainties. It is of utmost importance because the uncertainties in the measured data may affect the accuracy of fault diagnosis. The uncertainties are associated with the measurement, storage, and handling of oil samples [45]. It is crucial to note that the C_2H_6 gas has been used to replace C_2H_2 gas in DTM.

6. Duval pentagon method (DPM)

Duval Pentagons Method (DPM) has been proposed to resolve the issues which cannot be solved using DTM, Duval4, and Duval5. DPM consists of Duval Pentagons 1 (Pent1) and Duval Pentagons 2 (Pent2) using five key gases that constitute H_2, CH_4, C_2H_2, C_2H_4, and C_2H_6 in which they are positioned at the apices of the pentagon as shown in Fig. 7 [42,43,46,47].

Pent2 is used to further analyse the thermal fault (T1, T2 & T3) identified by Pent1 to understand the severity of paper carbonization involved in the fault, since such a suitable action can be taken to prevent disastrous damage [46,47]. Pent2 further defines the thermal fault areas as follows:

i O: overheating T <250 °C
ii C: thermal fault with paper carbonization

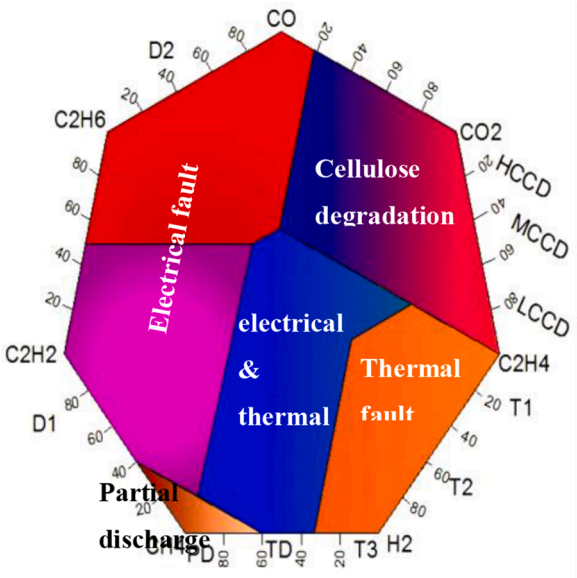


Fig. 10. Heptagon graph [50].

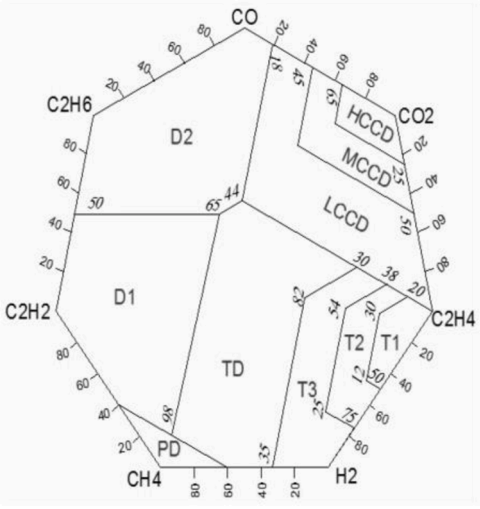


Fig. 11. Detailed Heptagon graph [50].

iii T3-H: high-temperature fault inside the oil

It is important to note that DTM, Duval4, Duval5, Pent1 and Pent2 are not correlated to each other. Therefore, different types of faults can be traced using these methods for the same set of DGA results, indicating the possibility of multiple faults occurring in the transformer [44,48]. The combination of Pent1 and Pent2 has been proposed in [47] to fully utilize both pentagons. Fig. 8 shows the overlap of both pentagons in which this Combined Duval Pentagon (CDP) Method simplifies the fault identification in the mineral oil-filled transformer. Table 5 describes the code definition to represent six areas related to thermal faults for Pent1, Pent2, and CDP.

Pattanadech [42] interpreted the dissolved gas in the transformer by using DPM to inspect the opened tank transformers in Thailand. Based on the gas concentration analysis, there were three major problems faced by the transformers as discussed in [42]. The proposed Duval pentagon method has reported 88% successful prediction and 88% consistency as compared to DRM with 77% and 71% respectively.

The Mansour Pentagon (MPM), a graphical technique in the shape of a pentagon that considers all five combustible gases, has been designed

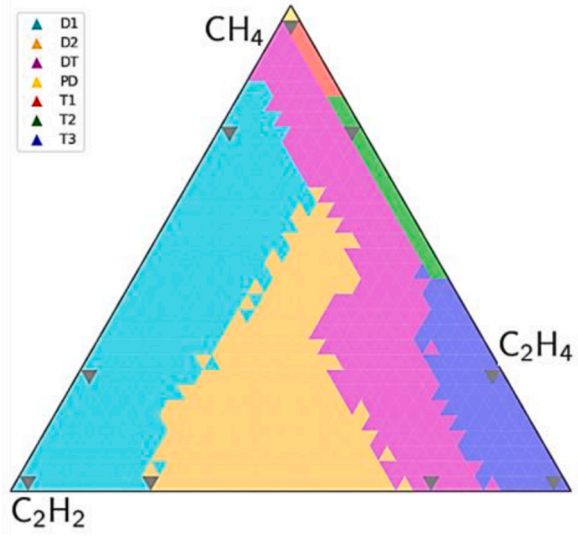


Fig. 12. DGA 4-Simplex [39].

[39,49] to improve the diagnostic accuracy presented by DTM. This is because DTM does not consider two combustible gases; C_2H_6 and H_2 which are the key gases for certain types of faults such as low overheating and partial discharge. Fig. 9 shows the MPM, with each pentagon representing the% concentration for each gas.

7. Heptagon graph method

The heptagon graph method is a new graphical representation to interpret DGA [50]. It uses seven types of gases which are methane (CH_4), acetylene (C_2H_2), ethylene (C_2H_4), ethane (C_2H_6), hydrogen (H_2), carbon dioxide (CO_2), and carbon monoxide (CO). The main significance of this method is the implementation of H_2 and C_2H_6 concentrations in improving the accuracy of diagnosis for specific types of transformer faults. Although other methods may neglect the formation of CO and CO_2, this method uses the generated CO and CO_2 from cellulose to differentiate transformer faults from cellulose insulation failure that is subject to transformer heating. Fig. 10 shows the heptagon shape that has been divided into six zones and each zone represents different types of faults in the power transformer. Six zones are represented by cellulose degradation and four types of faults, which are partial discharge, thermal fault, electrical fault, and electrical & thermal fault. Subsequently, the heptagon zones are further detailed to be more specific in identifying subcategories of faults as shown in Fig. 11.

The main drawback of this method is the difficulty in distinguishing the fault subcategories. For example, thermal faults can be classified into low, medium, and high thermal faults. Moreover, medium and high thermal faults consist of 2 different sets of percentages to differentiate the fault. For example, to determine the medium thermal fault, it can be one of these sets of percentages as follows:

a $50\% < C_2H_4 < 75\%$, 30% of H_2 and $25\% < CH_4 < 66\%$
b 50% of C_2H_4, $30\% < H_2 < 54\%$, and $25\% > CH_4$

Similarly, to partial discharge and electrical faults. Thus, it will make it more complicated which required to select a correct set of percentages. Otherwise, this will cause an error in determining the correct type of fault.

8. DGA 4-simplex method

The DGA 4-Simplex Method has been proposed in [39] to resolve the

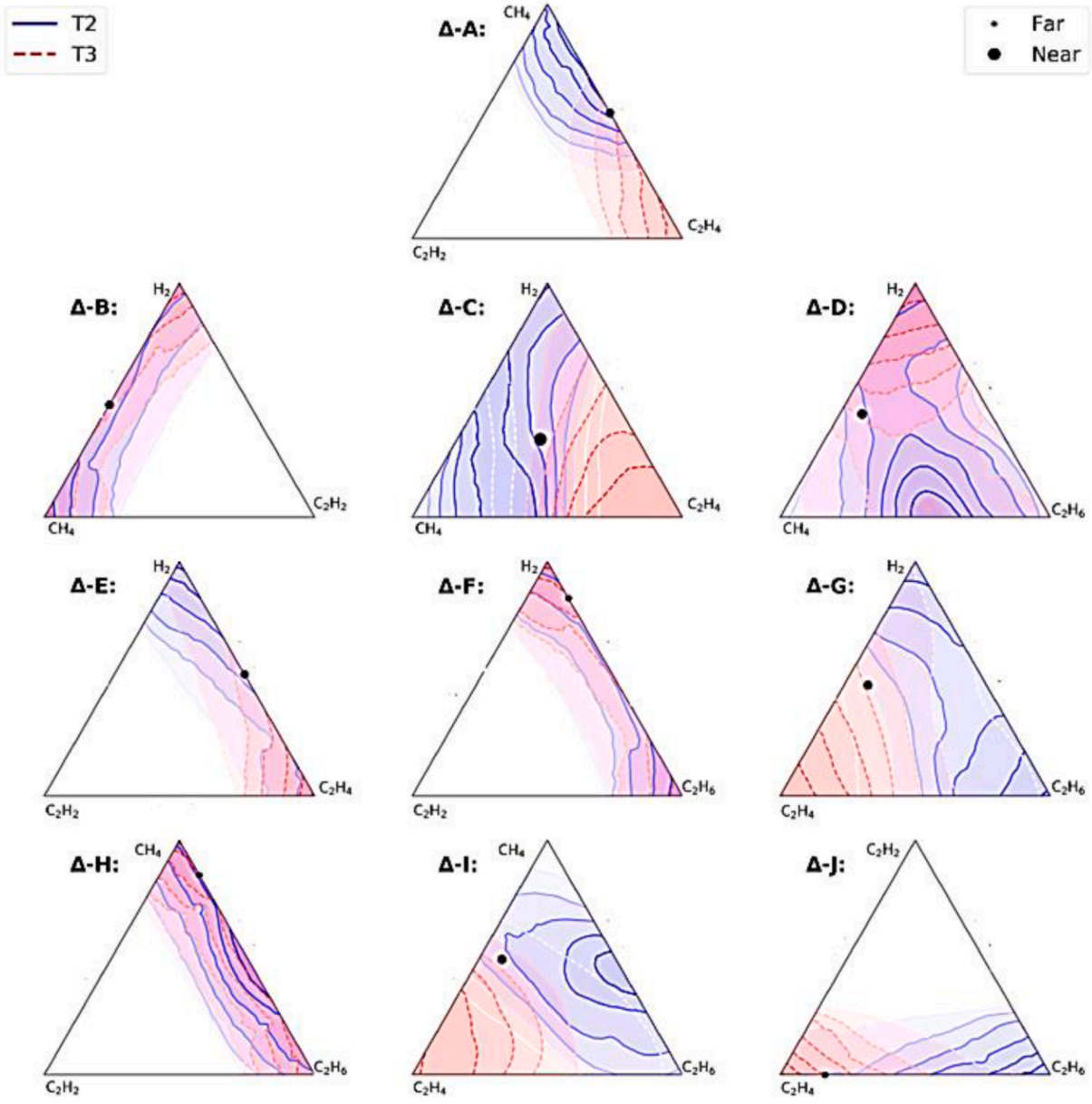


Fig. 13. Ten triangular faces of DGA 4-Simplex [39].

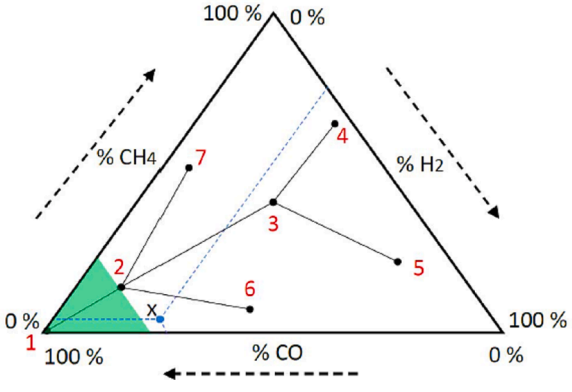


Fig. 14. Triangle representation of LEDT [51].

Table 6
Ratio between gases.

Ratio 1	Ratio 2	Ratio 3	Ratio 4	Ratio 5
$\frac{CH_4}{H_2}$	$\frac{C_2H_2}{C_2H_4}$	$\frac{C_2H_2}{CH_4}$	$\frac{C_2H_6}{C_2H_2}$	$\frac{C_2H_4}{C_2H_6}$

Table 7
Doernenburg Ratio Method [31].

Suggested Fault Diagnosis	Ratio 1	Ratio 2	Ratio 3	Ratio 4
Thermal Decomposition	> 1	< 0.75	< 0.3	> 0.4
Partial Discharge (Low-intensity PD)	< 0.1	Not significant	< 0.3	> 0.4
Arcing (High-intensity PD)	0.1 to 1	> 0.75	> 0.3	< 0.4

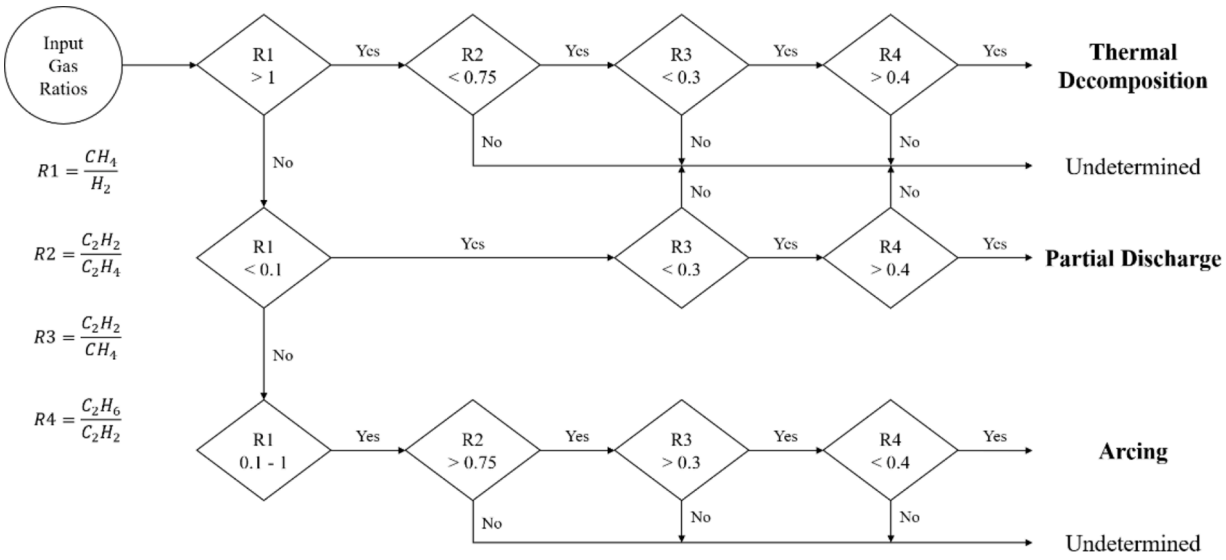


Fig. 15. Flow chart for DRM fault classification [31].

Table 8
Dissolved Gas Concentration (L1) for DRM [53].

Key Gas	L1 Concentration (ppm)
H_2	100
CH_4	120
CO	350
C_2H_2	35
C_2H_4	50
C_2H_6	65

Table 9
Subsets for six gas ratios [54].

Maximum	Minimum	Studying subsets	Ratio	Expression
	CH_4	1	R1	$\frac{CH_4 + C_2H_6}{H_2 + CH_4 + C_2H_6 + C_2H_4 + C_2H_2}$
	C_2H_6	2		
	C_2H_4	3	R2	$\frac{CH_4 + C_2H_4}{H_2 + CH_4 + C_2H_6 + C_2H_4 + C_2H_2}$
	C_2H_2	4		
	CH_4 & C_2H_6	5	R3	$\frac{C_2H_6}{CH_4 + C_2H_4}$
	CH_4 & C_2H_4	6		
	CH_4 & C_2H_2	7	R4	$\frac{CH_4 + H_2}{H_2 + CH_4 + C_2H_6 + C_2H_4 + C_2H_2}$
H_2	C2H6 & C_2H_4	8		
	C_2H_6 & C_2H_2	9	R5	$\frac{C_2H_4 + C_2H_2}{H_2 + CH_4 + C_2H_6 + C_2H_4 + C_2H_2}$
	C_2H_4 & C_2H_2	10		
	CH_4 & C_2H_6 & C_2H_4	11		
	CH_4 & C_2H_6 & C_2H_2	12	R6	$\frac{C_2H_2}{C_2H_4}$
	CH_4 & C_2H_4 & C_2H_2	13		
	C_2H_6 & C_2H_4 & C_2H_2	14		
	CH_4 & C_2H_6 & C_2H_4 & C_2H_2	15		

Table 10
Rogers Ratio Method [31].

Case	Suggested Fault Diagnosis	Ratio 2	Ratio 1	Ratio 5
0	Normal	< 0.1	0.1 to 1	< 1
1	Partial Discharge (Low energy density)	< 0.1	< 0.1	< 1
2	Arcing (High energy density)	0.1 to 3	0.1 to 1	> 3
3	Low temperature thermal	< 0.1	0.1 to 1	1 to 3
4	Thermal < 700 °C	< 0.1	> 1	1 to 3
5	Thermal > 700 °C	< 0.1	> 1	> 3

issue that occurs when using DPM to perform the DGA diagnostic interpretation. It was observed that DPM is restricted due to the plan affine coordinate system, in which it cannot be affinely independent. Thus, causing some gas combinations to be plotted at improper fault zones. As such, the triangle plots having affine independence can ensure unique barycentric coordinates for all points within the triangle thus allowing three variables to be defined in two-dimensional space. Significantly, allowing visualization of its interior to be projected onto its ten triangular faces. It is important to note that the points in the DGA 4-Simplex are represented by the combinations of the five key gases. In addition to that, the probability of each fault type at each point within the DGA 4-Simplex is determined using the Kernel Density Estimation. Fig. 12 shows the generalization triangle of the DGA 4-Simplex which is made up of its boundary and interior points, thus subsequently producing ten triangular faces as shown in Fig. 13.

The significant difference between DGA 4-Simplex and DTM is that the fault zones in DGA 4-Simplex are represented by four-dimensional probability distributions which are obtained from Kernel Density Estimation. Unfortunately, the main disadvantage of the DGA 4-Simplex is the difficulty to be visualized it. Additionally, a large and reliable database is required to train the DGA 4-Simplex. Nonetheless, the DGA 4-Simplex is more effective, accurate, and outperforms the other DGA interpretation methods.

9. Low energy degradation triangle (LEDT) method

The Low Energy Degradation Triangle (LEDT) Method has been inspired by the DTM except that the method incorporates H_2, CO, and CH_4 which are the dissolved gases with low energy into the triangular representation as shown in Fig. 14. It is important to note that transformer incipient faults usually begin as a low-energy deformation before evolving into a visible defect [16]. Therefore, it is essential to set up

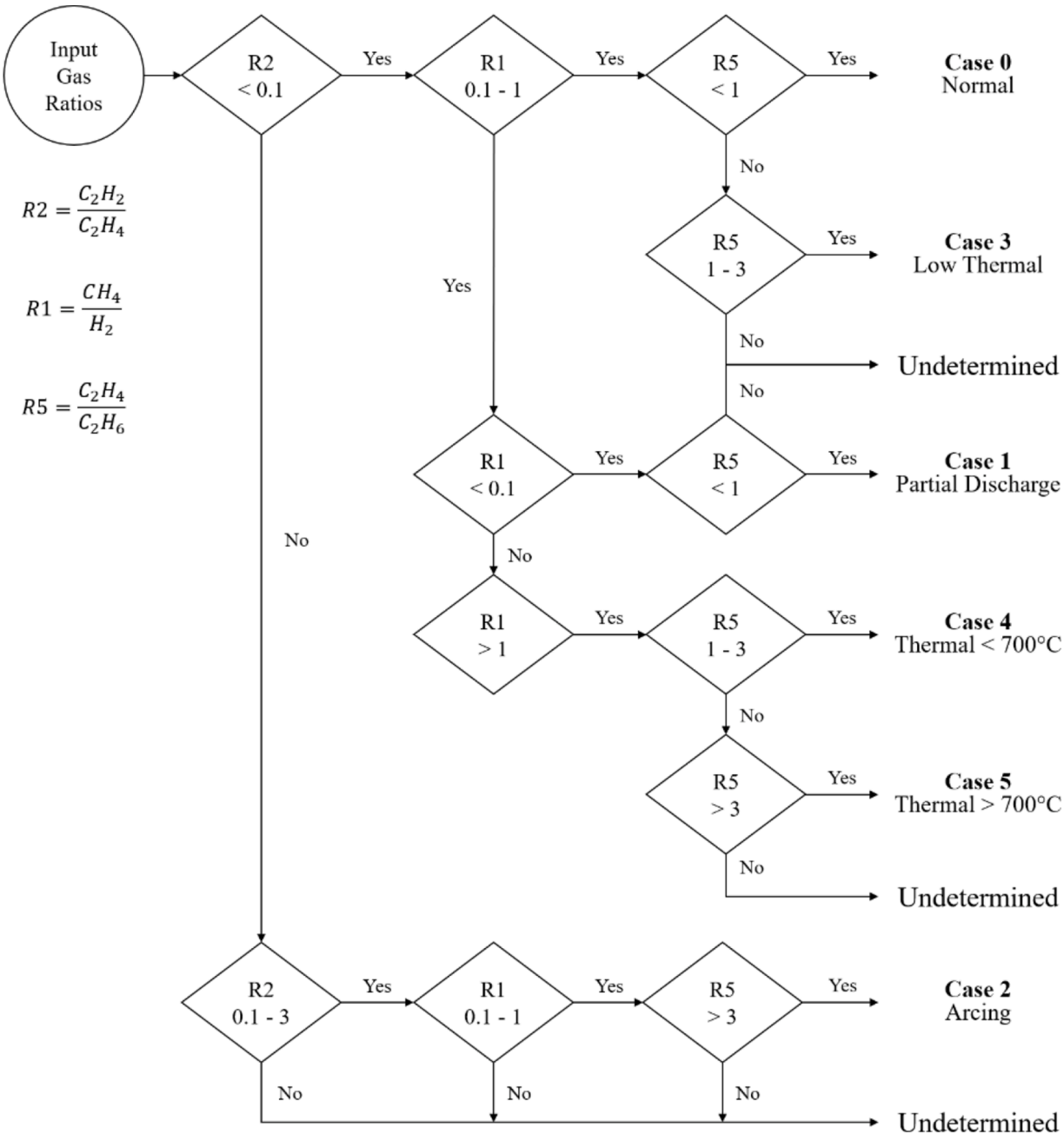


Fig. 16. Flowchart for RRM fault classification [34].

surveillance to check up on these changes even though everything appears to be normal. As a result, if the key gases associated with low energy conditions such as H_2, CH_4, and CO are taken into account, it can be a useful method for gaining an early warning. Thus, the proposed LEDT method can interpret the early state transition of the transformer's health status from normal to unhealthy conditions (internal faults).

Basically, a new transformer has trivial combustible dissolved gasses. However, after some intervals of operation with proximity to the rated temperature, it can be observed that most of the transformers begin to produce CO gas with insignificant levels of CH_4 or H_2 as illustrated in Fig. 14.

10. Ratio method

There are three methods that implement ratio-based calculations which are DRM, RRM, and IEC ratio methods. In this method, five gas ratios have been proposed, namely Ratio 1, Ratio 2, Ratio 3, Ratio 4, and Ratio 5 as shown in Table 6. It is important to note that DRM and RRM are based on IEEE Standard C57.104, whilst the IEC ratio method implements IEC 60,599 standard.

10.1. Doernenburg ratio method (DRM)

The Doernenburg Ratio Method (DRM) is the method based on IEEE Standard C57.104, utilizing four gas ratios which are Ratio 1, Ratio 2, Ratio 3, and Ratio 4. It has successfully diagnosed several fault conditions including thermal decomposition, partial discharge, and arcing, as shown in Table 7. Whereas Fig. 15 illustrates a flow chart for DRM fault classification.

In general, transformer faults are attributed to the breakdown of carbon and hydrogen bonds in hydrocarbon oil, which later forms the radical as end products. The radicals then recombine to form various gas

Table 11
IEC 60,599 ratio code [58].

	IEC 599	Ratio 2	Ratio 1	Ratio 5	
	Ratio characteristic gases	0	1	0	
	< 0.1	1	0	0	
	0.1 – 1	1	2	1	
	1 – 3	2	2	2	
	> 3				
Case No.	**Characteristic Fault**				**Typical Example**
0	No fault	0	0	0	• Normal ageing
1	Partial discharge of low energy density	0 but not significant	1	0	• Discharges in gas-filled cavities (due to cavitation, supersaturation, incomplete impregnation, or high humidity)
2	Partial discharge of high energy density	1	1	0	• Similar to Case 1 but with leading tracking or perforation of solid insulation.
3	Discharges of low energy (See Note 1)	1 – 2	1	1 – 2	• Continuous sparking in oil due to bad connections of different potential or floating potential. • Breakdown of oil between solid materials.
4	Discharges of high energy	1	0	2	• Discharges with power follow-through. • Arcing-breakdown of oil (between windings or coils, or between coils to earth) • Selector breaking current.
5	Thermal fault of low temperature (< 150 °C) (See Note 2)	0	0	1	• Local overheating of the core due to flux concentrations. • Increasing hot spot temperatures ✓ varying from small hot spots in the core
6	Thermal fault of low temperature (150 - 300 °C) (See Note 3)	0	2	0	✓ overheating of copper due to eddy currents ✓ bad contacts/joints (pyrolytic carbon formation) up to the core ✓ tank circulating currents.
7	Thermal fault of medium temperature (300 - 700 °C)	0	2	1	
8	Thermal fault of high temperature (> 700 °C) (See Note 4)	0	2	2	

Note 1:
For this event, the value for Ratio 2 tends to rise from 0.1 to above 3 and Ratio 5 from 1 – 3 to above 3 due to the spark developing in intensity.
Note 2:The value for Ratio 5 is defined by the gases that are mainly obtained from solid insulation decomposition.
Note 3:Fault condition is typically implied by the increment of gas concentrations. The value of Ratio 1 either below or above the unity value is reflected by many factors, including the design of the oil preservation system, the actual temperature level and the quality of the oil.
Note 4:
A higher value of C_2H_2 concentration implies the hot-point temperature is higher than 1000 °C.

which is dissolved in the transformer oil. The type of gas emitted depends on the type of hydrocarbon, the temperature of the nearby fault, and the duration of oil under electrical or thermal stress [11,52]. Compared to KGM, the ratio method such as DRM and Rogers Ratio Method (RRM) is a more accurate interpretation tool. However, RRM has a huge limitation in the gas ratio combination that does not fit any default conditions [17]. Apart from that, DRM diagnosis cannot be conducted if the key gas concentration of H_2, CO, CH_4, C_2H_2, C_2H_4, and C_2H_6 exceeds the relevant L1 concentration as shown in Table 8 [53].

Nanfak et al. [54] have applied both combinations of key gases and gas ratio methods to inspect the 475 samples dataset of the default transformer. Subsets were obtained from the decomposition of the main dataset that comprises the combination of maximum and minimum concentrations of the samples that can be referred to in [54]. Table 9 shows the obtained subsets based on six gas ratios that include five main hydrocarbons, namely H_2, CH_4, C_2H_2, C_2H_4, and C_2H_6. In each subset, different types of faults are determined by using the gas ratio method.

As shown in the table, 15 studying subsets are obtained from H_2 as the maximum concentration. It can be observed that since each subset is treated independently, it allows a higher range of ratios to be utilized to develop the submodel. Therefore, a total of 75 subsets can be developed from the main dataset. The obtained results show that the gas ratio method provides significant outcomes with a detection accuracy of more than 90% for PD, D2, and T3 types of faults. Besides that, more than 83% of dissolved gas was well diagnosed which is equal to 396 out of 475 samples.

10.2. Rogers ratio method (RRM)

Similar to DRM, RRM is also built on the IEEE Standard C57.104 but it only considers three ratios which are Ratio 1, Ratio 2, and Ratio 5. These ratios were selected based on the inspection experienced in the industry prescribing diagnostic cases as displayed in Table 10. Sun et al. described the effectiveness of this method to solve the abundance of failure investigation results for each case of gas analysis [55]. Moreover, RRM is utilized if the individual gas concentration exceeds the allowable limit. Thus, it does not require to have a minimum limit or specific gas concentrations to perform a valid diagnosis [12]. For example, RRM has been applied instead of DRM in detecting the presence of C_2H_2 since the specific concentration of C_2H_2 must be at least 1 ppm for DRM ratio diagnosis to be valid. Besides that, RRM is reported as better than DRM because it involved a broader combination of ratio ranges and thus resulted in a substantial reduction of no-fault or no-decision cases [34]. Furthermore, this method has been reported to be effective, since the results obtained are constant even after conducting a DGA diagnostic in each transformer oil fault case [56].

The significance of RRM is it links the findings of the failure investigation to the next results of the gas analysis as illustrated in Fig. 16. Whereas the main disadvantage of this method is it will give unreliable and inaccurate results for the testing samples with extremely high or low gas concentrations even though the fault is visible. For example, Taha et al. have reported that the RRM method has failed to explain the roots of the transformer fault [10]. Moreover, Almoallem et al. have reported that RRM generates out-of-code values for several DGA cases due to the

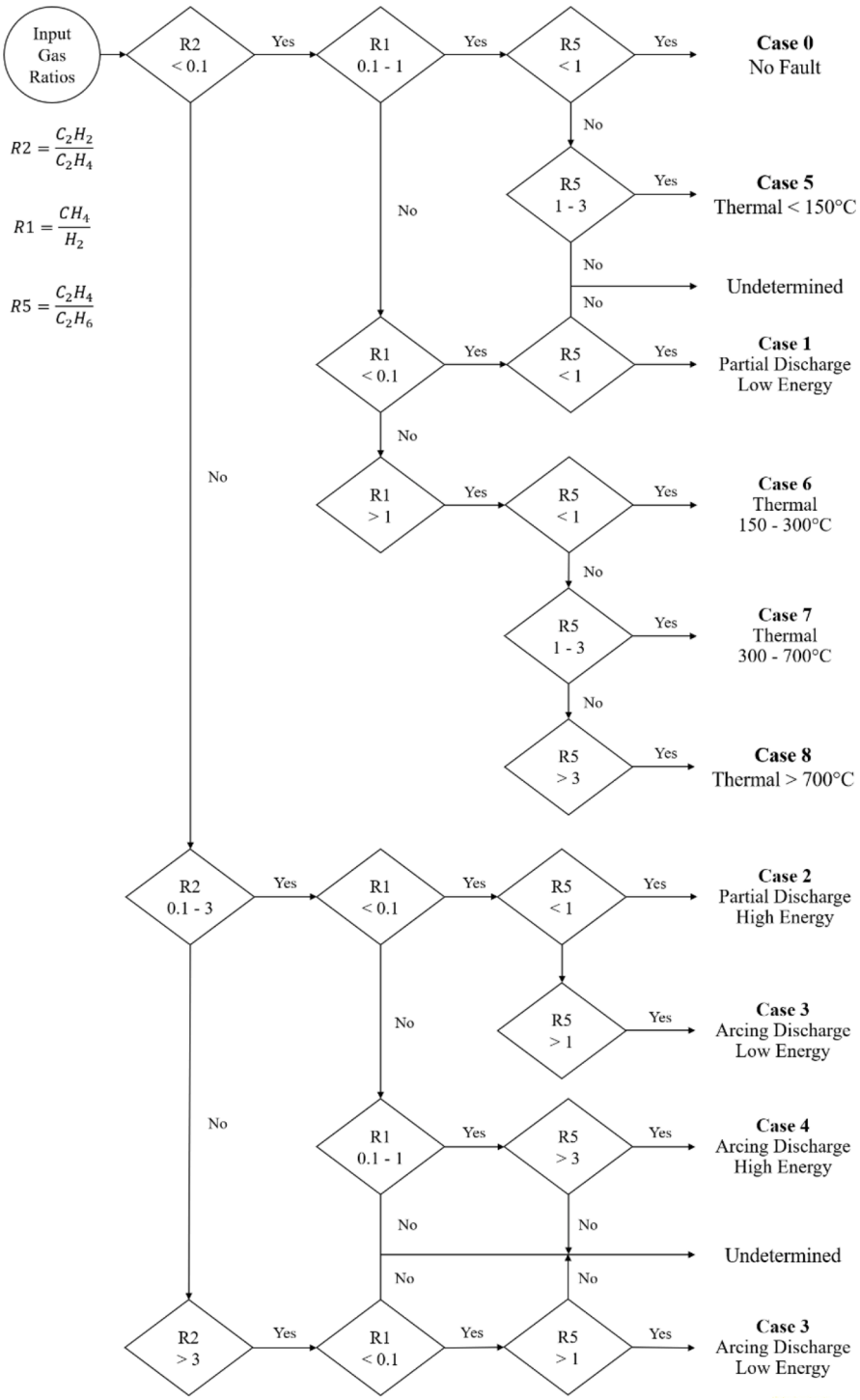


Fig. 17. Flowchart for IEC RM fault classification [58].

concentration of gases used in the ratios being insufficient [57].

10.3. IEC ratio method

The IEC ratio method (IEC RM) has a similar approach to RRM in that it uses only three gas ratios to diagnose faults such as thermal faults in different temperature ranges (300 – 700 °C). There are four detected electrical faults such as normal ageing, partial discharge, discharge of low and high energy cases, as well as the thermal fault of varying temperatures, as shown in Table 11. Compared to the other DGA methods, IEC RM utilized the IEC 60,599 standard instead of IEEE Standard C57.104. Fig. 17 illustrates a flowchart for the IEC RM that includes different types of faults and ratio limits. However, the current IEC RM is not accurate whereby in some cases, the method cannot interpret the fault that occurred, and consequently, the result is categorized as 'Undetermined'.

Taha et al. have proposed a modified IEC RM which can be implemented as new diagnostic rules to allow it to adapt to the actual fault type in the 'Undetermined' category [13]. The significance of the proposed modified IEC code is that it can overcome the 'Undetermined' state, is able to diagnose many cases, including failed cases, and enhance the accuracy of current IEC RM. Besides that, Ward et al. have claimed

Table 12
Different types of faults categorized by different DGA methods.

Fault No-Fault	KGM	DTM	Duval4 & 5	DRM	RRM Normal	IEC No-fault
Thermal	Thermal oil	Thermal < 300 °C	Stray gas	Thermal decomposition	Low thermal	Thermal < 150 °C
		Thermal 300 - 700 °C	Paper carbonization ($T > 300$ °C) Overheating (< 250 °C)		Thermal < 700 °C	Thermal 150 - 300 °C
	Thermal cellulose		Thermal 300 - 700 °C			Thermal 300 - 700 °C
		Thermal > 700 °C	Thermal > 700 °C		Thermal > 700 °C	Thermal > 700 °C
Partial Discharge	Partial discharge	Partial discharge	Partial discharge	Partial discharge	Partial discharge	Low energy density High energy density
Arcing	Arcing	Low energy discharge High energy discharge		Arcing	Arcing	Low energy discharge High energy discharge
Mix Thermal- Electrical		Mix thermal- electrical				

Table 13
Advantages and disadvantages.

	Advantage	Disadvantage
Key Gases	1 Simple	1 Fault-based on the highest number of gas concentration 2 Very conservative
DRM	1 Accuracy is satisfactory 2 Accurate judgement for thermal fault	1 Incomplete coding – some cases cannot be diagnosed 2 Limited range of fault
RRM	1 Wide range of thermal fault	1 Incomplete coding – some cases cannot be diagnosed 2 Low accurate judgement for discharge fault
IEC	1 Wide range of faults 2 The arrangement of the fault sequence is more reasonable starting from incipient fault to severe fault	1 Incomplete coding – some cases cannot be diagnosed
DTM	1 Easy visualization 2 Accurate judgement for overheating fault	1 Mistakes can easily occur. 2 Cannot differentiate whether the transformer has a fault or not. 3 Lower accuracy for discharge fault
Duval4 & 5	1 Gives specific thermal faults at different temperature ranges	1 Unable to apply for electrical fault
Duval Pentagon	1 Better prediction	1 The coordinate system is not dependable
4simplex	1 Effective 2 Accurate	1 Difficult visualization 2 Required large data
LEDT	1 Effective for analysis of low molecular weight gases	1 Cannot be applied to high molecular-weight gases

that the modified IEC RM has the flexibility to adjust the limits of the gas ratio and its connection in order to engage with more correct fault types identification [1].

11. Correlation faults between all methods

Table 12 summarizes the ability of each conventional DGA method to discriminate between different types of faults. As shown in the table, it can be observed that the same type of fault can have different characteristics which have been differentiated with more detail. For instance, partial discharge can be more elaborated to low and high energy densities as described in the IEC method. Similarly, arcing faults can be categorized as low- and high-energy discharges, as mentioned in the IEC method and DTM. Whereas Table 13 summarizes the advantages and disadvantages of each method.

12. The comparison in accuracy

Each method shows different accuracy and consistency under the prediction of DGA in normal conditions. The percentages of successful prediction and consistency are calculated as follows:

$$S_{Fn} = \frac{R_{Fn}}{No.of\ cases\ Fn} \times 100 \tag{8}$$

$$C_{Fn} = \frac{\int_{n=1}^{n=5} S_{Fn}}{No.\ of\ fault\ types} \tag{9}$$

where F_n is a type of fault code, n is 1,2,3,4, etc. The accuracy of predictions is classified into two types which are (i) the accuracy of the predicted cases only and (ii) the accuracy based on the total number of cases. Muhamad et al. [59] have stated that the IEC Ratio Method has given the highest accuracy of predicted fault cases which is above 92%, followed by 91% in the RRM and 90% in the DRM. Meanwhile, the DTM has recorded the highest accuracy of the predicted total cases which is around 89%, followed by the KGM. Unfortunately, both RRM and DRM methods have recorded the lowest accuracy on the total cases prediction which are 45% and 41%, respectively.

Guoda et al. [60] have reported that the DTM has the highest accuracy over 77% in total predictions, followed by 66% using DRM and 56% using IEC Ratio Method. Meanwhile, in [61], it has stated that the DTM has the highest number of correct diagnoses which is above 96%. The DRM showed the lowest number of incorrect diagnoses which is below 3% but with 26% of unresolved diagnoses. Unfortunately, KGM showed the lowest percentage of correct diagnoses, which is below 42%, indicating the failure of the system especially in distinguishing thermal faults in oil and cellulose.

13. Conclusion and future works

DGA technology is a reliable monitoring tool for identifying various types of faults occurring in transformers based on insulation oil. However, the issue of its accuracy is still beyond 100%, and as such there is always room for improvement. On the basis of the literature, the reliability performance of each DGA method is observed and compared to each other. It can be observed that each DGA method does not give 100% accuracy, but through observation, the incorrect result ensured by any DGA method can be overcome by the other DGA method. Thus, it can highlight the probability of hybrid the DGA method to increase the accuracy. Besides that, it is possible to introduce a new comprehensive approach by modifying the parameters and frameworks from the existing method. Moreover, the effect of an ageing transformer on insulation oil could also be performed to investigate more thoroughly the factors that contribute to the inaccuracy of the DGA method. In addition, more on-site data is required to obtain a comprehensive analysis of each DGA method, besides improving the reliability and accuracy of the method. Furthermore, the involvement of experienced engineers added benefit in verifying the transformer cause of the fault concerning the insulation oil. Therefore, this paper presents detailed information on the merits and demerits of each DGA method for future research. It serves as a reference and guide to determine the most suitable DGA method for performing detailed diagnostics associated with a specific type of transformer fault.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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